Data Analyst vs **Data Scientist** vs **AI Engineer**

Steps involved in the **AI/ML Pipeline**

- Step1 : Data Collection
- Step2 : Data Cleaning & Preprocessing
- Step3 : Exploratory Data Analysis (EDA)
- Step4 : Feature Engineering
- Step5 : Model Selection & Training
- Step6 : Model Evaluation
- Step7 : Model Deployment

Skills for different Career Paths

1. Data Analyst

- Data Collection
  - SQL
  - Excel / Google Sheets
- Data Cleaning & Preprocessing
  - Python
  - Statistics
- EDA
  - Power BI
  - Tableau

2. Data Scientist

- Data Collection
  - SQL

- Data Cleaning & Preprocessing
    - Advanced Python
    - NumPy & Pandas

- EDA
    - Matplotlib
    - Seaborn

- Feature Engineering

- Model Selection & Training
    - Scikit-learn
        - Supervised ML
        - Unsupervised ML
        - Reinforcement Learning
    - Git & Github (additional)
    - Math
        - Statistics
        - Linear Algebra
        - Probability
        - Calculus

- Model Evaluation

3. AI Engineer

- Data Collection
    - SQL

- Data Cleaning & Preprocessing
    - Advanced Python
    - NumPy & Pandas

- EDA

    - Matplotlib
    - Seaborn

- Feature Engineering

- Model Selection & Training
    - Scikit-learn
        - Supervised ML
        - Unsupervised ML
        - Reinforcement Learning
    - Git & Github (additional)
    - Math
        - Statistics
        - Linear Algebra
        - Probability
        - Calculus
    - Deep Learning (PyTorch or TensorFlow)
    - ANN
        - FNN
        - RNN
        - CNN
        - Transformer etc.

- Model Evaluation

- Model Deployment
    - Basic Deployment
    - API Building
    - DSA (optional, asked by some companies)